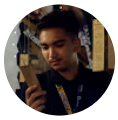
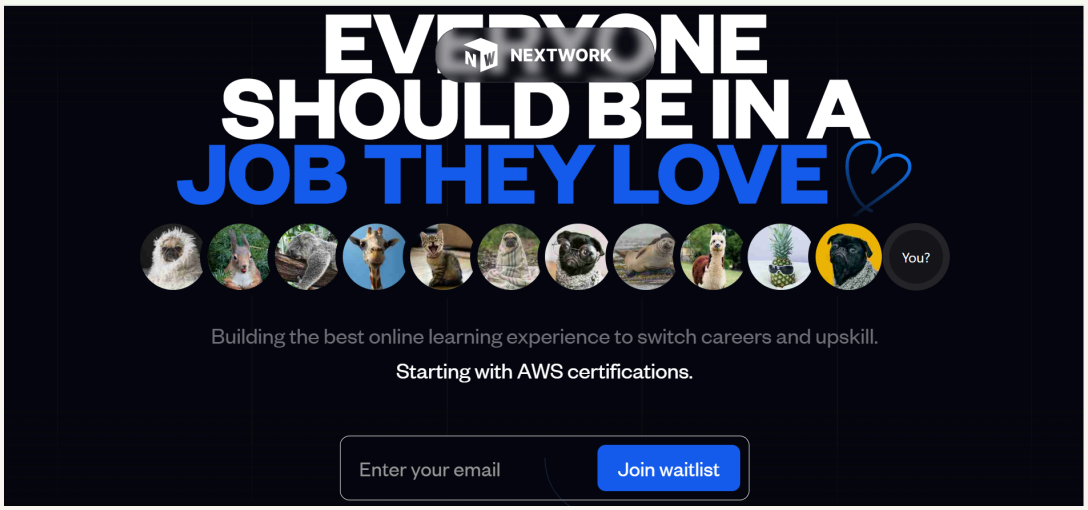
Host a Website on Amazon S3



RAMOS II JUANITO



Introducing Today's Project!

Project overview

In this project, I will demonstrate how to create an Amazon Simple Storage Service (S3) bucket to store files and images needed for a website deployment. I am doing this project to learn how Amazon S3 works and to gain an understanding of how it can store and retrieve any amount of data at any time from anywhere on the web.

Tools and concepts

This project uses Amazon S3 as the primary service. Key concepts covered include deploying a static website, enabling public access, and managing permissions through Access Control List (ACL) configuration.

Time, challenges, and wins

This project took approximately 45 minutes to complete. The most challenging part was navigating the AWS Management Console to configure ACL settings correctly in order to make the website publicly accessible and avoid the 403 Forbidden error. The most rewarding part was successfully making the website publicly available through proper configuration.

How I Set Up an S3 Bucket

What I did in this step

In this step, I will open Amazon S3 because we need to create a storage space for our website files.

How long it took to create the bucket

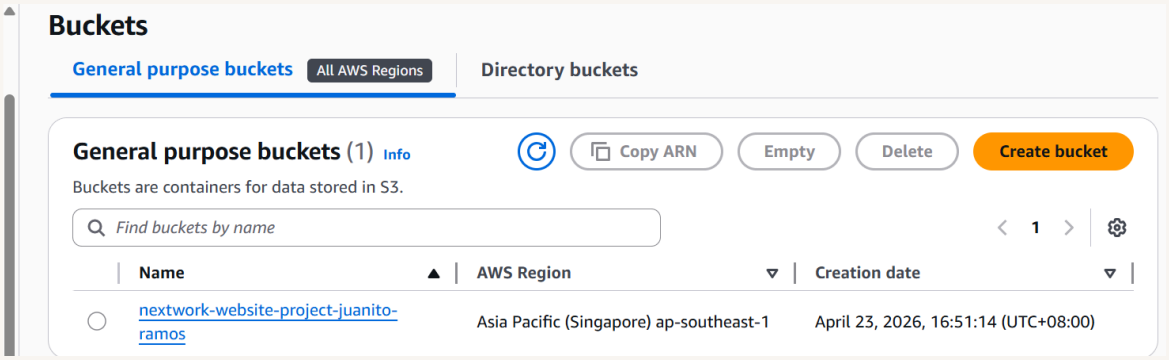
Creating an S3 bucket took me 15 minutes.

Region selection

The region selected for the S3 bucket is Asia Pacific (Singapore) (ap-southeast-1), as it is the closest available AWS region to the Philippines.

Understanding bucket name uniqueness

S3 bucket names must be globally unique. This means that once a bucket name is created, no other AWS account can use the same name.


Buckets
General purpose buckets
All AWS Regions
Directory buckets
General purpose buckets (1) Info
Buckets are containers for data stored in S3.
Copy ARN
Empty
Delete
Create bucket
Find buckets by name
1
Name
AWS Region
Creation date
nextwork-website-project-juanito-ramos
Asia Pacific (Singapore) ap-southeast-1
April 23, 2026, 16:51:14 (UTC+08:00)

Upload Website Files to S3

What I did in this step

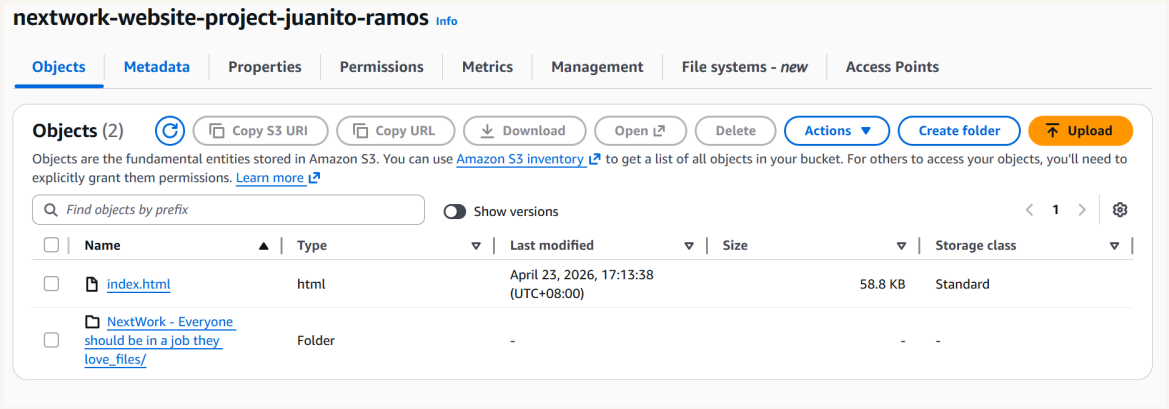
In this step, the HTML file for the website and a ZIP file containing the required images are downloaded. These files will be uploaded to the S3 bucket in a later step.

Files I uploaded

I uploaded two items to my S3 bucket: the index.html file and a folder containing 44 image files.

How the files work together

These two are related because the index.html file serves as the main webpage, while the folder contains the images used within the website. The HTML file references these images to properly display the website’s content and layout. Both are necessary because the website would not display correctly without the image assets, and the images alone would not function without the HTML file to structure and present them.


nextwork-website-project-juanito-ramos Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - new
Access Points
Objects (2)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Show versions
1
Name
Type
Last modified
Size
Storage class
index.html
html
April 23, 2026, 17:13:38 (UTC+08:00)
58.8 KB
Standard
NextWork - Everyone should be in a job they love_files/
Folder
-
-
-

Static Website Hosting on S3

What I did in this step

In this step, I will configure the S3 bucket for static website hosting and verify whether the public website endpoint is accessible.

Understanding website hosting

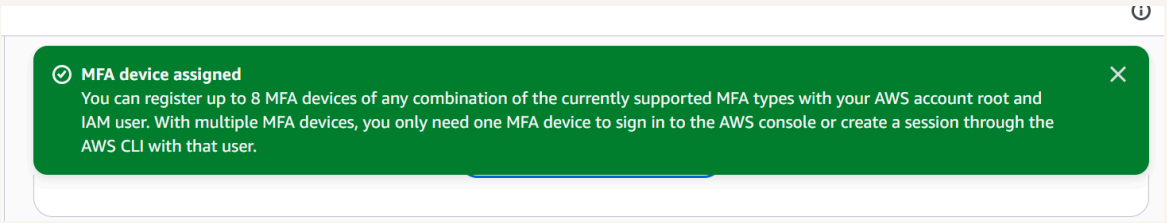
Website hosting refers to the process of making a website publicly available on the internet so that users can access it through a web browser.

How I enabled website hosting

To enable website hosting with my S3 bucket, I configured the static website hosting settings and specified the index.html file as the default document.

Access Control Lists (ACLs)

An Access Control List (ACL) is a mechanism used to define the level of access granted to specific users for particular resources. In this project, ACLs were enabled to allow access permissions to be shared with other AWS accounts. Access to the S3 bucket and its objects can be managed and specified using ACLs.


MFA device assigned
You can register up to 8 MFA devices of any combination of the currently supported MFA types with your AWS account root and IAM user. With multiple MFA devices, you only need one MFA device to sign in to the AWS console or create a session through the AWS CLI with that user.

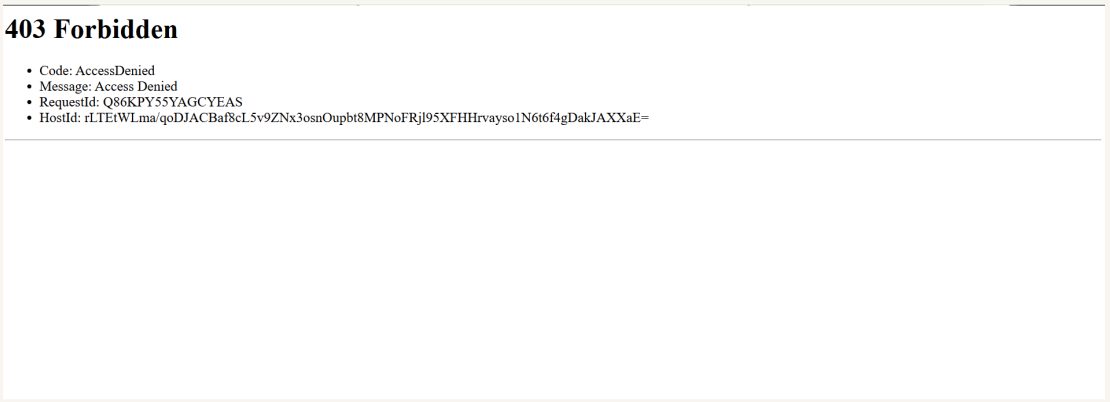
Bucket Endpoints

Understanding bucket endpoint URLs

Once static website hosting is enabled, S3 generates a bucket endpoint URL, which is the public web address used to access the hosted website.

What I saw when I tested the endpoint

When I first visited the bucket endpoint URL, I saw a 403 Forbidden error. The reason for this error was that the bucket and its objects were not yet configured for public access.



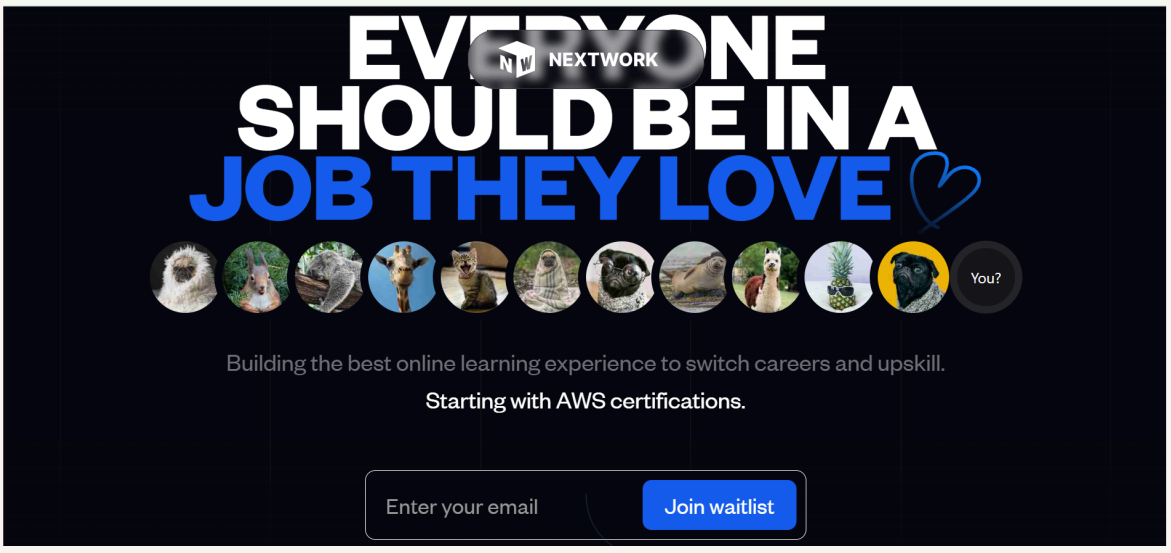
Success!

What I did in this step

In this step, I configured the S3 bucket and its objects to be publicly accessible so that the website can be accessed over the internet.

How I resolved the 403 error

To resolve this 403 Forbidden error, I updated the S3 bucket permissions to allow public access to the website files.



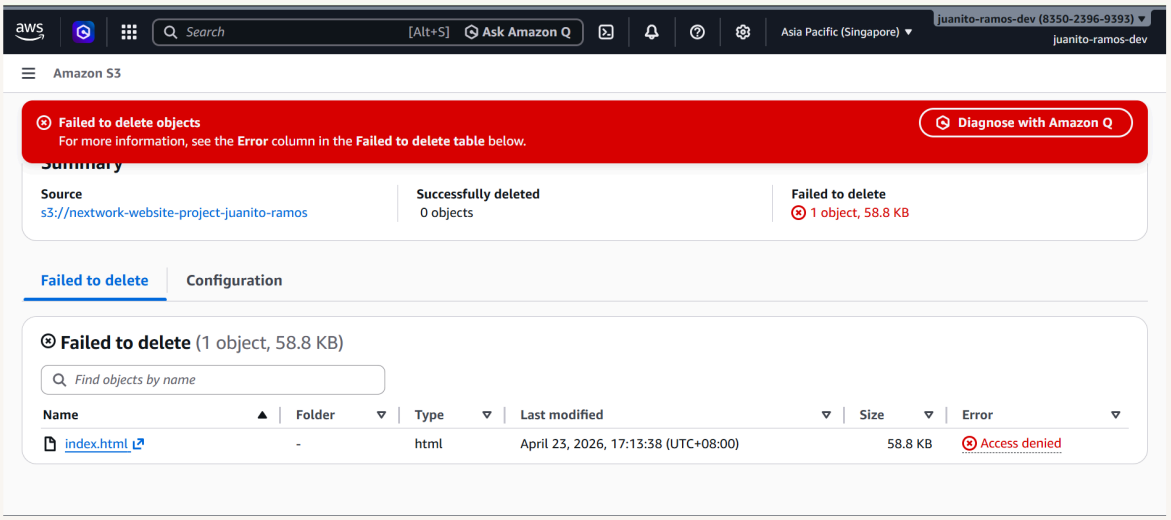
Bucket Policies

What I did in this extension

In this project extension, I will set up a bucket policy to prevent users from deleting the index.html file. The purpose of this is to learn how to use S3 bucket policies to control access and protect critical files at the root of a static website.

Understanding bucket policies

An alternative to ACLs are bucket policies, which are JSON-based rules used to define access permissions for an S3 bucket and its objects. The benefit of using bucket policies is that they provide more flexibility and centralized control over access management, allowing you to define detailed and scalable permission rules. ACLs, on the other hand, are useful for simpler, object-level access control and for granting basic permissions to other AWS accounts.



What my bucket policy does

My bucket policy was configured to restrict deletion of the index.html file. I tested this by attempting to delete the file using the S3 console, and I saw that the action was denied due to the policy restrictions.

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